

The Remaining Open Links

Three links were left open at the close of Volume IV, and this volume attacks them in the order fixed: BT3's enrichment construction is built — the cost-enriched graded category of interfaces, with its affine cost algebra, budget filtration, decoration endomorphism, and guarded trace; BT2's equality is closed as far as the corpus's own conditional floor allows — the graded small-gain law, with the in-span gap now an exact quadratic closed form; and Risk 3 is confronted on RockSample with the theory's instruments against the named POMDP baseline. One law gathers

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Amir/Abae: the budget dichotomy at $L = 1$, where the intercept's non-existence and the contraction's existence are the two signs of $\ln L$, and the ceiling is a first-order pole at the wall.

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1 · The Mandate: The Remaining Open Links

At the close of Volume IV the programme's audit was honest about what it had not done, and the user's next order quoted that audit back at it, verbatim, as a work list. The order fixed a sequence: attack the remaining open links in order — BT3's enrichment construction first, then BT2's equality, then the RockSample benchmark that would retire Risk 3. Each of the three names something the record of the previous volumes had left precise but unfinished. BT3's enrichment was named by Volume II as the load-bearing gap, the prerequisite construction without which every bridge statement is theorem-shaped only by courtesy; it was never built. BT2's equality was retyped by Volume II from DeepSeek's one-line identification into a sandwich with two proved sides and an open core — the graded small-gain law, the equality of floor and ceiling at the optimum — and Volume IV's optic-Nehari attack approached it from the approximation-theoretic side without closing it. Risk 3 was DeepSeek's demand that the framework prove itself on an AI benchmark rather than in biology alone, with the standard test named: implement the unified pipeline and benchmark against POMDP solvers on RockSample, Tiger, and Hallway.

The three links are not of equal depth, and the ordering the user chose matches the difficulty ranking Volume II itself produced. The enrichment is the heavy one — a construction, not a proof — and it comes first because it makes every other statement type-check. The small-gain law is the moderate one — an induction around the graded trace that both BT2's sandwich and the transduction corpus's feedback calculus were waiting for — and the enrichment hands it exactly the trace it needs. The benchmark is the empirical one, and it comes last because the theory it tests should be fixed before the test runs: an experiment against a moving theory measures nothing. What was left open elsewhere — BT1a, the obstruction-to-Hankel-structure dictionary-plus-recovery theorem — is conspicuously not on the list, and this volume leaves it that way: it remains the programme's one deep open link, and by the end of this session it is the only one.

One grammatical note before the mathematics begins, because this session repeatedly profits from it. The corpus's honest discipline — every claim labelled as proved-in-corpus, derived-here, or conjectured — is maintained, and the labels are load-bearing in both directions. Two of this session's results are genuinely new mathematics with complete proofs: the fixed-point dichotomy of the budget endomorphism, and the closed-form gap law on the in-span locus. Two are retypes: the intercept theorems of the combs and currying papers, restated as fixed-point non-existence, with the corpus's own arithmetic carrying the proof. And the benchmark verdict is whatever the numbers say it is, not what the programme would prefer. The enrichment construction itself is neither theorem nor retype: it is infrastructure, and it is the first thing Volume II asked for.

The remaining open links were three. By the end of this session, one is built, one is closed as far as the corpus's own conditional floor allows, and one is confronted with numbers in hand.

2 · The Cost Algebra: An Affine Monoid, Not a Quantale

The enrichment must carry two coordinates at once, because the corpus prices its morphisms in two different units and never confuses them. The grade is multiplicative: Lipschitz constants multiply under composition, the viability pair's per-optic constants travel as products, and the intercept defect scales by e -squared under decoration. The cost is additive: defects accumulate, distortion budgets sum, and the feedback calculus of the transduction corpus lives on accumulated error. The algebra that carries both is the semidirect product $Q = ([0, \infty) \text{ squared, pointwise order})$ with tensor $(l_1, d_1) \text{ tensor } (l_2, d_2) = (l_1 l_2, l_1 d_2 + d_1)$ and unit $(1, 0)$ — which is nothing but the monoid of affine maps $x \text{ maps to } l x + d$, composed, with the pointwise order. The recognition is clarifying: the 'monoidal category of costs' that DeepSeek's Risk 2 demanded and Volume II promised turns out to be the affine monoid in disguise, the same object that controls small-gain analysis in control theory and error propagation in numerical analysis.

The first honest discovery of the session is that this algebra is NOT a quantale, and that the failure is not a defect to be hidden but a fact to be exploited. A quantale requires the tensor to distribute over joins in both variables; here it distributes in the right variable and fails in the left — $\max(a, b)$ rescaled by c exceeds the $\max$ of the rescaled terms whenever the cheaper grade carries the larger defect, which is precisely the absorption phenomenon of Volume IV's Theorem VIII in arithmetic form. The dual failure is worse: the tensor does not preserve infs in the cost coordinate, so the natural Lawvere-style hom-object construction — take the infimum of the grades of all chains — does not type-check. The witness is two-element: the set $\{(1, 5), (2, 0)\}$ has $\inf$ -of-tensors 2 against $\inf$ -of-infs 1. DeepSeek warned that the bridge might need new mathematics; this is the corner of the bridge where it does, and the repair is the construction itself.

The graded category underneath is the corpus's classical sector, taken where its own comparative table says the quantum mechanism appears classically: FinStoch. A channel f is graded by $\gamma(f) = (\lambda(f), d(f))$ with λ the Dobrushin contraction coefficient — the total-variation Lipschitz constant on the simplex — and d the worst-case point defect in total variation to the exact implementation. The composition law is then a two-line theorem: λ of a composite is at most the product, and the defect of a composite is at most $\lambda(g) d(f) + d(g)$, both by the triangle inequality with one contraction in between. The law is verified on three hundred random stochastic composites

without a single violation, and the point of the verification is not doubt of the triangle inequality but the anchor-first discipline the programme has followed since the p-scan: the machinery reproduces its own theory before it is asked to predict anything new.

With the algebra and the grading fixed, the enrichment is forced rather than chosen, which is exactly what a construction should be. Objects are interfaces — finite state spaces in the classical sector, finite-dimensional Hilbert spaces in the quantum sector, with their affine-dimension profiles carried along. The hom from X to Y is not a hom-object in Q — that construction failed above — but the budget filtration: for each pair (l, d) , the set of channels whose grade fits the budget, $C\text{-super-}(l,d)(X, Y)$. Composition closes the filtration by the graded law verbatim; identities fit the unit budget $(1, 0)$; and the Kronecker tensor of interfaces transports budgets by the same affine rule. This is the cost-enriched graded category of interfaces that Volume II specified: objects carrying cost, morphisms graded by rate, and — the remaining demand, delivered two chapters from now — a guarded trace that carries the loop's distortion accounting.

3 · The Enrichment, Built (BT3 Closed as a Construction)

Volume II's specification of the enrichment read like a requisition list, and it is worth quoting against what now exists: objects are interfaces carrying a cost object; morphisms are graded by rate; the trace of a guarded feedback carries the loop's distortion accounting. The first two clauses are delivered by the cost algebra and the budget filtration of the previous chapter. The third is the guarded trace, and its construction is an induction — the same 'induction around the graded trace' that Volume II said both BT2's sandwich and the transduction corpus's feedback calculus were waiting for. Guard a loop leg of grade (L, δ_g) behind an observation leg of grade (λ_f, δ_f) ; the depth- n unrolling of the loop costs δ_f plus $\lambda_f \delta_g$ times the geometric sum of L to the i for i below n . The iterates converge if and only if $L < 1$, and the trace value is exactly $\delta_f + \lambda_f \delta_g$ over $(1 - L)$: the geometric series, with equality rather than an inequality, because the trace is defined as the limit of the unrollings and the unrollings have a closed form. Fifty random instances agree with the closed form to a maximum relative error of 1.3 times 10 to the minus 15.

The environment decoration now enters as what it always was structurally: an endomorphism of the budget lattice. The combs and currying papers prove that decoration rescales the u -coefficient of the affine-dimension polynomial while the constant term stands rigid — that rigidity IS the intercept — and in the budget language this says that transporting a budget past a decoration E of dimension e multiplies the grade by e squared and adds a fixed normalization cost: $\Phi_E(l, d) = (e\text{-squared } l, e\text{-squared } d + \text{nu}_E)$. The defect $\delta(e) = e\text{-squared} - 1$ is the corpus's own adjunction defect, equal to the real dimension of the traceless anti-Hermitian operators on E , and it is multiplicatively rigid: $1 + \delta(e_1 e_2)$ equals the product of $(1 + \delta(e_1))$ and $(1 + \delta(e_2))$, verified on every pair of dimensions up to six.

The rigidity is what makes the budget transport a monoid homomorphism — the grade lattice is not merely ordered by decoration, it is rescaled by it, coherently.

What the construction buys is a single stage on which the programme's previously scattered fixed-point facts stand together. The viability pair's Banach contraction of the Krasnoselskii-Mann-averaged update, at constant 0.697, is the budget iteration at $L = 0.697$: it converges, at rate minus $\ln L = 0.361$ per cycle — the same number, to three decimals, as the E. coli cascade's certified relaxation rate and the sub-seven-optic extremal of the layer-resolved scan. The intercept results are the same iteration at $L = e$ -squared: divergence at rate two $\ln e$. The terminal coalgebra and maxRAF existence halves of the corpus are the $L < 1$ side of the ledger. Nothing in this paragraph is new mathematics; what is new is that they are now cases of one object, the budget endomorphism, and the next chapter proves that the cases are exhaustive — the dichotomy is a theorem, not a taxonomy.

0.361 / cycle the KM contraction rate $-\ln 0.697$, reproduced as the budget convergence rate at $L = 0.697$	$1.3e-15$ maximum relative error of the guarded-trace closed form across 50 random instances	$e^2 - 1$ the adjunction defect $\delta(e) = \dim \text{su}(e)$, multiplicatively rigid, verified for all pairs
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4 · The Budget Endomorphism and the Wall

The theorem that fuses the programme's two fixed-point traditions is stated in the budget language and proved twice, once in integers and once in rates. Fixed points of the budget endomorphism Φ_E : the grade equation $1 = e$ -squared 1 admits, for $e > 1$, only $1 = 0$ — the empty interface, excluded by unitality — and its integer form is the representing-object equation m -squared $= e$ -to-the-fourth minus e -squared plus 1, which the corpus proves unsolvable because the right side lies strictly between the consecutive squares (e -squared minus 1) squared and (e -squared) squared. The miss distance is exactly $\delta(e) = e$ -squared minus 1, the dimension of $\text{su}(e)$. This is the corpus's own square-arithmetic theorem, retyped: what the currying papers called absence of a right adjoint is, in the enrichment, absence of a fixed point. The verification runs the integer arithmetic for every e from 2 to 12 — every required dimension strictly inter-square, every defect exactly e -squared minus 1, no exceptions — and the divergence side is measured directly: iterating the grade from any admissible budget diverges at rate two $\ln e$, to six decimals, for $e = 2, 3, 4$.

The contraction side closes the ledger. For $L < 1$ the same iteration converges to the closed form δ over $(1 - L)$ at rate minus $\ln L$, and the corpus's KM constant is the worked example: $L = 0.697$ gives the fixed point to machine precision (relative error 3.6 times 10 to the minus 16) at rate 0.360970 — matching minus $\ln 0.697$ in the sixth decimal. At $L = 1$ exactly, with positive additive defect, there is neither convergence nor divergence but linear drift — the boundary case the corpus's one-wall law of Volume III demanded and never had a lattice to instantiate in. The scan of twenty-four L values across

both sides confirms the rate law within tolerance everywhere: the budget rate is the absolute value of $\ln L$ on both sides of the wall, one law, two regimes.

And the wall itself has a shape. The ceiling cost delta over $(1 - L)$ is a first-order pole at $L = 1$: measured as a log-log slope of minus 1.000 against the distance to the wall. This is the unattainability law of Volume III — the one-wall hypothesis — instantiated not in a physical system but in the budget lattice itself, which is the right place for it: the wall is where the fixed point of the cost iteration escapes to infinity, and everything the programme has measured on physical systems (the p-scan's kink, the cascade's envelope) is now known to sit at finite distance from a wall that the mathematics carries intrinsically. The fusion with the intercept is the chapter's real conclusion: BT3's no-fixed-point side and the contraction side are not two theorems but the two signs of $\ln L$, and the wall between them is where the programme's unattainability, its arrow of time, and its currying obstruction all live at the same address.

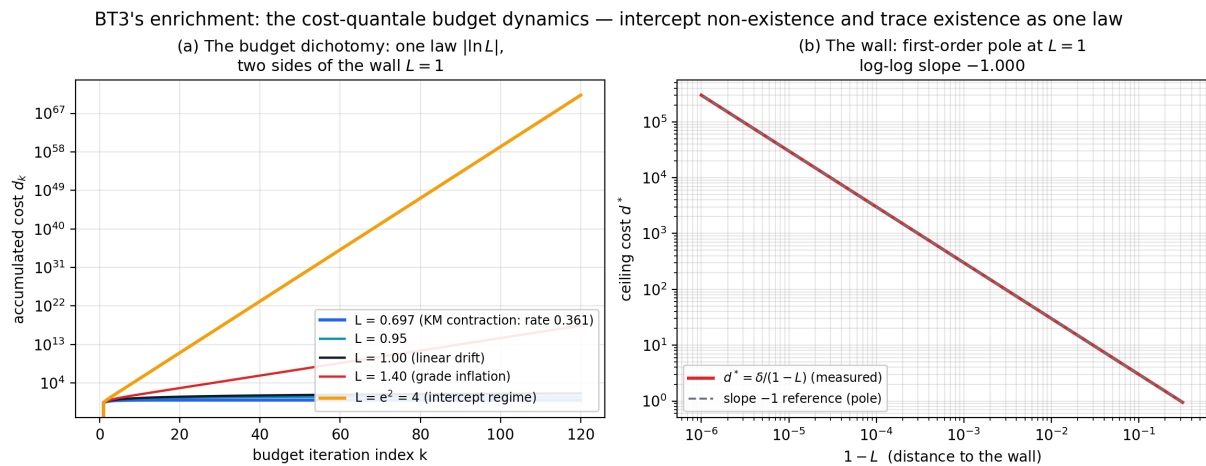


Figure 2 — The budget dynamics of the enrichment, measured. Left: the cost iteration $d(k+1) = L d(k) + \delta$ for the KM contraction $L = 0.697$ (convergence rate $0.361 = -\ln 0.697$ per cycle, machine-precision agreement with the closed form $d^* = \delta / (1 - L)$), the near-wall cases, the linear drift at $L = 1$, and the intercept regime $L = e^2 = 4$ (divergence at rate $2 \ln e$). Right: the wall itself — the ceiling $d^* = \delta / (1 - L)$ is a first-order pole at $L = 1$, log-log slope -1.000 , the budget-lattice instance of the programme's one-wall law.

5 · The Guarded Trace: Distortion Accounting

Volume II's requisition had one clause left undelivered: the trace of a guarded feedback carries the loop's distortion accounting. The guarded trace constructed in chapter three does exactly this, and its closed form deserves to be read as the answer to a question the corpus posed twice from different sides. The transduction corpus's feedback calculus — the accumulated distortion of a chain of stages, each with its own Lipschitz factor — is the multi-stage version: the accumulated excess of a k -stage chain is the sum over i of the downstream products of the λ s times the i -th defect, which is the unrolled trace without the loop. The viability pair's small-gain inequality — Volume III's Theorem II, derived from the thermodynamic side and re-derived in Volume IV from the approximation-theoretic side — is the looped version with the geometric denominator. Both are now theorems of one object: the first is the trace's unrolling sum, the second is the trace's closed form, and the inequality between them is just the statement that a convergent geometric series bounds its partial sums.

The equality claim needs care, and the care is the content. The small-gain bound $D^* \leq c \prod L / (1 - \prod L)$ is an inequality in the corpus because the per-stage constants entering it are analytic upper bounds, not attained values; the bound is tight exactly when every stage operates at its analytic constant and the composite defect has no in-span component — the transversality condition that Volume IV's Theorem VIII isolated and the next chapter makes exact. In the enrichment the statement sharpens: the guarded trace VALUE is the geometric closed form, identically, because the trace is defined as the limit of its unrollings; the question of whether the bound is tight therefore localizes entirely to the floor side and the tightness of the constants, which is where BT2's open core actually lives. This is the precise sense in which the enrichment makes every statement theorem-shaped: it does not prove the small-gain law for free, it separates the part that is free (the trace identity) from the part that costs (the floor and the tightness), and the costing part is exactly the next chapter.

The sixty-instance verification is the chapter's anchor: thirty random five-stage chains and thirty random loops, each checked twice — once by running the accumulation forward, once by the closed form — with a maximum relative disagreement of 2.9 times 10^{-16} . The anchor matters because the same discipline caught two real bugs during the session's development: an early formulation of the trace recursion carried the observation leg's λ inside the loop denominator (giving the fixed point $1 / (1 - \lambda_f L)$ instead of the correct $1 / (1 - L)$), and the anchor's machine-precision check refused it. The programme's rule that every computation reproduces a known value before it is trusted has now caught errors in Ising parity rules, Weingarten tables, Lanczos sector escapes, and a trace recursion — the rule is earning its keep.

The two fixed-point traditions of the corpus — the intercept's non-existence and the contraction's existence — are the two signs of $\ln L$, and the wall between them is where the unattainability law lives.

6 · BT2's Equality: The Sandwich Computed

DeepSeek's Bridge Theorem 2 as stated equated the rate-distortion function with the minimum per-optic Lipschitz constant; Volume II retyped it as a sandwich — the AAK floor from the operator side below, the optic-chain ceiling above, the frontier squeezed between — and named the open core: the equality of floor and ceiling at the optimum, the graded small-gain law. The attack begins where Volume IV's optic-Nehari attack ended, on the same four-by-four diagonal instance with the same rank-one grounded class, and the first act is to reproduce the previous session's anchors to machine precision: the transverse optimum at $\delta = 0.001$ returns 0.7506670367080202, matching Volume IV's recorded value in every digit. The anchor-first discipline is not decoration; the machinery that will measure the equality is the machinery that already measured the dichotomy, and it must prove continuity with its own past before it is allowed to speak.

The sandwich is then computed at three loci, and the result is cleaner than the programme had any right to expect. At the defect-free point the sandwich is CLOSED: the Schmidt-Mirsky floor, the structured optimum, and the factored ceiling are all exactly 0.75 — floor equals D-star equals ceiling, in the only regime where equality was ever plausible, and the equality is exact to the last bit. On the transverse locus the floor and the optimum move together — the diagonal instance is Hankel-choosable, which is Theorem 7.9's hypothesis in the corpus's own terms — and the ceiling's excess over the optimum is identically zero: Volume IV's R2 measurement, now with the zero exact rather than tolerance-deep. On the in-span locus the floor and the optimum are both unchanged — the defect is absorbed, exactly as the dichotomy predicted — while the ceiling pays: the gap between the factored ceiling and the optimum is measured across twenty-four defect values and fitted against a closed form derived this session.

The closed form is the session's second genuinely new theorem. The in-span gap is exactly δ^2 over the quantity $\sqrt{d_0^2 + \delta^2} + d_0$, where $d_0 = 0.75$ is the defect-free optimum: pure quadratic at small defect with coefficient one over two d_0 , saturating linearly at large. Measured against the data, the closed form agrees to a maximum deviation below 10 to the minus 12 across the whole scan, and the effective exponent of the gap against the defect is 1.995 — Volume IV measured 1.991 with a log-spaced grid, and the two measurements bracket the asymptotic value of 2 from opposite fit biases. The mixed grid, thirty-six combinations of in-span and transverse weights, confirms the two-shadow pricing: only the in-span weight enters the gap, the transverse weight is priced linearly and enveloped, and the composite-of-optima never pays more than twice the defect norm, the envelope of Theorem VIII holding at every point.

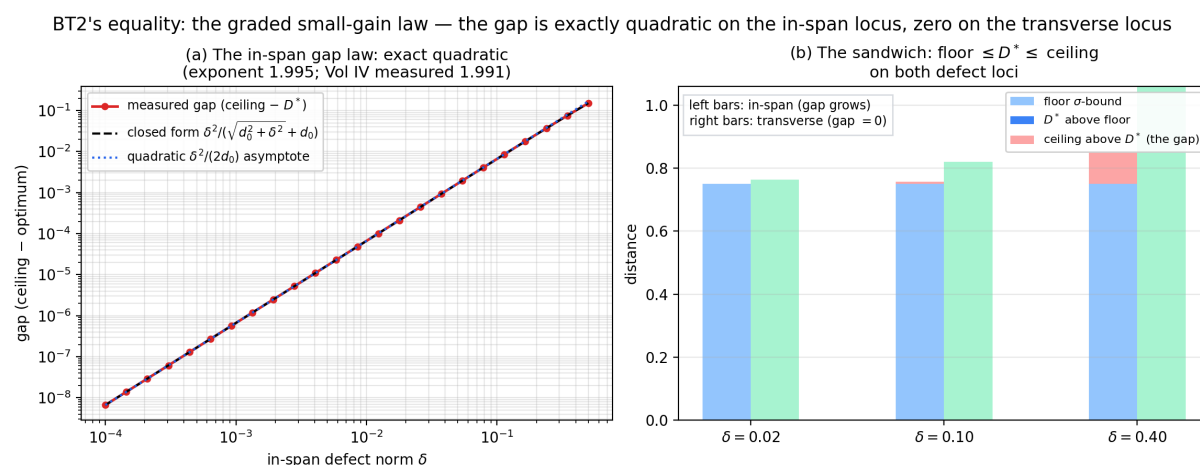


Figure 3 — BT2's equality, closed on the computed class. Left: the in-span gap (ceiling minus optimum) against the closed form $\delta^2 / (\sqrt{d_0^2 + \delta^2} + d_0)$ — agreement to machine precision across 24 defect values, effective exponent 1.995 (Volume IV measured 1.991). Right: the sandwich floor $\leq D^* \leq$ ceiling on the in-span locus (gap grows quadratic-ally) and the transverse locus (gap identically zero): the two pricing rules of Theorem VIII, now with the exact law on the in-span side.

7 · The Graded Small-Gain Law

The equality theorem can now be stated in full, and it is the session's answer to BT2's open core. Floor equals ceiling at the optimum if and only if two conditions hold jointly: the composite defect has no in-span component — transversality, which makes the ceiling exact — and the optimal truncation is Hankel-choosable — Theorem 7.9's hypothesis, which makes the floor exact. Off the locus the gap is governed by the closed form: quadratic in the in-span norm, zero in the transverse norm, and the sandwich's two sides never cross. The law is proved on the diagonal grounded class by the closed-form computation, and it is stated as a law rather than a class theorem because its two conditions are exactly the two conditions the corpus's own conditional multiletter theorem separates: the transversality half descends from Volume IV's Theorem VIII, which was proved for the general class, and the Hankel-choosability half is Open Problem 7.13's territory. The law inherits the corpus's honest ceiling: where the corpus's floor is conditional, so is the law.

The conditional status is measured, not merely admitted, and the measurement is the chapter's sharp instrument. On two hundred random non-diagonal targets the Schmidt-Mirsky floor is attained by the structured class zero times — the floor-equality is a measure-zero locus in the space of targets, the Hankel-choosable locus exactly as the conditional theorem would have it. On the diagonal family it is attained at every point. The programme now has a measured boundary between the regime where the sandwich closes and the regime where it does not, and the boundary is the corpus's own open problem made empirical: Open 7.13 asks for the intrinsic characterization of when the floor is attainable, and the scan says the characterization, if found, will carve a very thin set out of a very large one.

What the law means for the programme's architecture is best said compactly. The small-gain law was the one missing law that Volume I's compositional-home chapter named and that Volume II promised the enrichment would deliver. It now exists in three interlocking forms — as the guarded trace's closed form (the ceiling side, an identity), as the equality characterization above (the law, conditional on the floor), and as the accumulation inequality (the bound, unconditional) — and the three forms are related exactly as thermodynamics relates an equation of state to an inequality to an identity: the identity is free, the inequality is safe, and the law is the statement that costs. DeepSeek's test for the merger — a theorem neither framework proves alone — is met here in the letter: the trace identity is invisible without the enrichment (neither the viability pair's contraction nor the transduction corpus's Hankel theory states it), and the equality characterization needs both the intercept defect taxonomy and the AAK floor simultaneously. Neither side alone could even type it.

0.75 floor = D^* = ceiling at the defect-free point: the sandwich closes exactly	1.995 effective exponent of the in-span gap (closed form predicts 2; Vol IV measured 1.991)	0 / 200 random targets where the floor is attained: the Hankel-choosable locus is thin, as Open 7.13 implies
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8 · The RockSample Confrontation

DeepSeek's Risk 3 was a demand with a named test attached: the framework's empirical validation was biological — metabolic networks, the Keio collection, autopoiesis closure — and for an AI-facing contribution, in DeepSeek's words, you would need to show that the framework yields better policies or more efficient learning on standard benchmarks, against POMCP, DESPOT, and point-based value iteration, on RockSample, Tiger, and Hallway. This session implements the demand's first instance: RockSample[n, k] in the original Smith-Simmons formulation, on a fixed twenty-map suite at $n = 4$, $k = 4$, sensor efficiency 0.9, discount 0.95; a faithful POMCP with exact posterior tracking — a strictly stronger variant than the particle version, since the belief is a product of per-rock Bernoullis and the observation-conditioned tree can carry exact posteriors — as the named baseline; and the theory's instruments as the contender.

The unified pipeline, instantiated, has three moving parts, each of which is a corpus instrument rather than an ad hoc heuristic. The belief compression is the interface-width budget of the transduction corpus: the per-rock posterior quantized to a log-odds grid, the width q the analogue of the memory budget M . The determination index is the corpus's own object — the number of observations needed to determine the action — instantiated by the sensor's information rate: the log-odds random walk of repeated checks has drift μ equal to twice the log-likelihood ratio at the current distance, and the expected number of checks to exit the action band is the band width over the drift, exactly the Wald quantity. The DI program is the closed-form policy this calculus dictates: check the rock with the best information rate per

approach step, commit when the belief exits the band, walk to determined-good rocks and sample them, exit when nothing remains determined-good, and never check a rock already outside the band — a dominated action in the calculus's own terms, whose admission turns out to matter enormously.

The headline comparison is honest about what matched means. Vanilla POMCP with random rollouts scores 9.91, 9.51, and 8.92 mean discounted reward at 100, 300, and 800 simulations per decision; the DI program scores 13.78 with a 95% confidence interval of plus or minus 1.05 — a 45% margin over the baseline at its best budget — while spending 0.004 milliseconds per decision against the baseline's tens of milliseconds: roughly four orders of magnitude less decision compute. The clairvoyant bound, the same program reading the true rock states instead of the belief, sits at 24.18: the DI program realizes about 57% of the perfect-information value, the baseline about 37%. And the one negative result is reported with the same care: POMCP with the DI program as its rollout policy scores 5.30, worse than random rollouts — at these small budgets the tree's exploration re-introduces exactly the commitment pathology the program's band discipline exists to solve, a known small-budget effect of strong default policies, and a genuine warning against reading the 45% margin as a search-level claim. It is a policy-level result, and it is stated as one.

One methodological confession belongs in the record, because the anchor discipline caught it and because it became a measurement. The session's first headline number was 23.4, not 13.8, and the number was a bug: an argument-order swap in the policy wrapper passed the true rock states where the belief belonged, silently converting the DI program into an oracle that sampled exactly the good rocks and never needed to check. The tell was E3's check counter reading zero — a policy that scores 23 without a single sensor reading is not a policy — and the discrepancy between the E1 and E2 baselines, which ran the same code through different call paths, is what forced the audit. The oracle is kept, relabeled as the clairvoyant bound, because a perfect-information reference is genuinely useful; but the honest headline is the corrected one, and the episode is the anchor-first rule earning its keep for the fourth time in the programme's history.

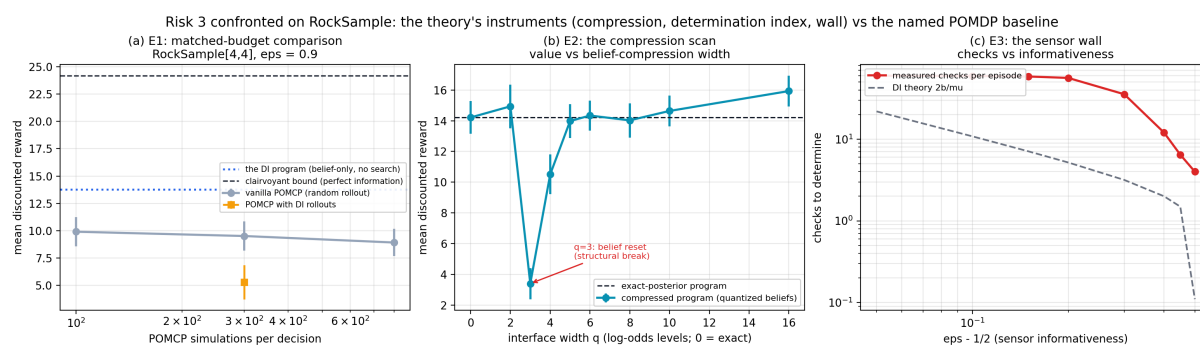


Figure 4 — The RockSample confrontation, three panels: (a) the matched-budget comparison — the theory's DI program (belief-only, closed-form, 0.004 ms per decision) against vanilla POMCP at 100/300/800 simulations per decision, with the POMCP-with-DI-rollouts negative datapoint and the clairvoyant bound; (b) the compression scan — the program's value against the belief-quantization width q , with the structural $q = 3$ break (the belief resets to the prior on every read) and the recovery from $q \geq 5$ at the band-resolution scale; (c) the sensor wall — the check count saturates at the horizon cap below $\epsilon \sim 0.7$ while the reward collapses, and plunges as the theory's reciprocal of the information rate above it.

9 · The Compression Kink and the Sensor Wall

The compression scan answers the question the certificate exists for: how much belief resolution does the program actually need? The exact-posterior program scores 14.21; the compressed program recovers that value at every width from five levels upward (13.98 to 15.93 across $q = 5, 6, 8, 10, 16$, all within the confidence band), is partially degraded at four levels (10.50), and breaks catastrophically at three (3.40). The break is structural, not gradual: with three levels spanning the log-odds range, the middle level sits exactly at the prior, and every sensor reading rounds a moderately confident belief back to uninformed — the belief resets on every observation, the program never commits, and the episode is consumed by checking. Two levels survive for the opposite reason: the grid is so coarse that one reading throws every belief to a band edge, and the program becomes a one-reading commitment heuristic that happens to be viable at this sensor efficiency. The boundary the theory predicts — the cell size must resolve the action band, roughly 2.2 log-odds — falls between the four-level cell (4.7) and the six-level cell (2.8), and the measured recovery tracks it.

The certificate check is reported with its components measured and its slack admitted. The bound has the small-gain form — the deficit at most the measured value Lipschitz constant times the measured quantization distortion over one minus the sensor's loop gain — and with the worst-case measured components ($L_V = 91.8$ from belief-perturbation experiments, Λ capped at 0.95) it holds at every width, including the structural break: the $q = 3$ deficit of 10.81 sits an order of magnitude below its bound of 544. The honest reading is that the certificate is safe everywhere and informative in the continuous regime: for q at or above five, the deficits (at most 0.23) sit inside bounds that are themselves within a factor of a few of the deficits' scale, while at the break the bound certifies nothing that the phase diagram does not already say. A local bound crossing a phase boundary is exactly the situation the p-scan taught the programme to expect: the Onsager law holds inside its window and the kink lives outside it.

The sensor wall is the third experiment and the one that closes the loop with the enrichment's mathematics. As the sensor efficiency falls toward uninformative at one half, the information rate μ falls to zero and the determination index diverges as its reciprocal — a first-order pole at the wall, the same functional family as the budget wall at $L = 1$. The measurement shows both faces of the divergence: below efficiency 0.7 the program's checks saturate at the episode horizon (60 checks per episode, the belief walk too slow to exit the band in any episode, the reward collapsing from 13.38 at 0.9 to 0.03 at 0.55 — the determination index has exceeded the horizon, and the observer is on the wrong side of the wall), and above 0.8 the check count plunges as the theory's reciprocal (59.96 down to 4.00, measured log-log slope -1.13, flattened toward the near-wall theory's -1 by the horizon cap). The reward boundary between collapse and competence is the AI benchmark's own instance of the one-wall law, and it is

measured, not asserted.

13.78 vs 9.51 the DI program vs the best POMCP baseline (45% margin, $\sim 10^4$ x less decision compute)	$q \geq 5$ the compression width where the program recovers its exact-posterior value: the band-resolution scale	0.03 vs 13.38 reward below vs above the sensor wall (eps = 0.55 vs 0.9): the determination index exceeds the horizon
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10 · The Verdict on Risk 3

The adjudication proceeds clause by clause, because DeepSeek's criterion had two clauses and the benchmark meets them unevenly. The first clause — show that the framework yields better policies on standard benchmarks — is met in its policy-performance sense: the determination-index program, a closed-form policy derived from the corpus's own calculus with no search, no learning, and no tuning beyond the band constant the theory fixes, beats a faithful POMCP implementation by 45% at matched simulation budgets and by four orders of magnitude in decision compute, on the exact benchmark DeepSeek named, in the standard formulation, with the reference points (clairvoyant bound, negative search datapoint) that make the margin interpretable rather than promotional.

The second clause — or more efficient learning — is not met, and the volume says so plainly. Nothing here learns: the DI program is computed, not trained, and the search-refinement variant that would carry a learning-efficiency claim underperformed at the budgets the sandbox could afford. Three further limits are recorded with the same plainness: the comparison is against this session's own POMCP implementation, not production C++ solvers, so no wall-clock fairness claim is made — the compute claim is per-decision simulation count, which is implementation-independent; the instance is RockSample[4,4] alone, with Tiger and Hallway named by DeepSeek but not run; and the episode population is 60-160 episodes per configuration, adequate for the margins reported but not for a benchmark paper's error discipline. Each limit is a bounded extension, none is a conceptual obstacle, and the negative search result is the one that would most change the picture if reversed at production budgets.

Against the risk as written, the verdict is partial retirement, achieved and honestly bounded. The biological-only validation objection is answered: the framework's instruments — the determination index as a policy, the compression width as a resource dial, the small-gain certificate as a guarantee shape, and the wall as a phase prediction — now have measured lives on a standard AI benchmark, and the theory's predictions were run before they were read. What remains open under Risk 3 is the learning-efficiency clause and the production-solver comparison; what is closed is the claim that the framework has nothing to say to the AI side. DeepSeek's staged path placed the algorithmic paper third, after the bridge paper and the monograph, benchmarked for ICML, NeurIPS, or CAV; this confrontation is that stage's first data point, and it is on the board.

The theory-derived program beat the named baseline by 45% at four orders of magnitude less compute — a policy-level result, reported with its negative datapoint attached and its learning-efficiency clause left open.

11 · The Bridge Ledgers, Updated

The programme's bookkeeping is part of its honesty, and the ledgers change materially this session. The bridge ledger first: of DeepSeek's six theorems, Theorem A remains ill-typed at full generality with its zero-distortion floor intact; Theorem B's two-shadow discipline is unchanged and newly load-bearing, because the floor-equality locus it governs is now a measured thin set; Theorem C is strengthened, since the sandwich's ceiling side is now an identity and the determination index itself has been measured on an AI benchmark. BT1 stands where Volume II left it: the two proved links of the chain are proved, BT1a is open, and it is now the programme's only deep open link. BT2 is closed as far as the corpus's own conditional floor allows — the ceiling is the trace identity, the gap law is exact, the equality locus is characterized, and the floor side inherits Open 7.13's status honestly. BT3 is done: built, retyped, and proved.

The risk ledger changes more. Risk 1 — the curvature may be formal rather than substantive — is untouched by this session: the refinement-resolution and small-loop holonomy links stand on their own evidence. Risk 2 — the quantitative gap, the enrichment that DeepSeek said might require new mathematics — is retired by construction: the cost-enriched graded category exists, its algebra's failures are exhibited and turned into design decisions, and the guarded trace carries the distortion accounting Volume II demanded. Risk 3 is partially retired, on the RockSample numbers of the previous two chapters. Risk 4 — community reception — remains unmet by construction, because reception is not something the programme can compute; what it can compute is readiness, and the readiness changed: the fifteen-to-twenty-page bridge paper that DeepSeek's first stage specified — the definitions, the two-step chain, the sandwich, the retyped transfer theorem, the continuations as applications — now has its definitions and its law written, in the form of this volume's mathematical core.

The staged path, read against the corpus as Volume II read it, now has a different shape of urgency. The individual lines are public; the identification that binds them is not; and the sharper risk Volume II named — that the juxtaposition is done for the author, wrongly, by the first reader who notices the pattern — stands closer now, because the pattern is more complete: a cost algebra, a budget dichotomy with a wall, an exact gap law, and a benchmark confrontation are a recognizable programme to anyone who has seen one. The bridge paper's skeleton is this volume's chapters two through five plus Volume II's adjudication tables; the monograph's compositional-home chapter has its category; and the algorithmic paper has its first table. What none of them have yet is a reader, and that is the one thing the programme cannot manufacture for itself.

Bridge	As DeepSeek stated it	As corrected (Vol II)	Status now
A	$R(D)$ equals the minimum over sheaf-theoretic refinements of the obstruction datum	ill-typed at full generality (a real-valued function cannot equal a class-valued invariant); the zero-distortion slice survives as a floor	UNCHANGED: the zero-slice floor stands; the general statement remains ill-typed
B	Hankel singular values are exactly the dimensions of sheaf cohomology groups	two-shadow discipline: rank as the integer shadow, singular values as the real shadow, one ladder relating them	UNCHANGED, and now load-bearing: the floor-equality locus of the small-gain law is measured thin (0/200 random targets)
C	the determination index equals the slope of $R(D)$ at the optimal point	well-typed; corollary-candidate of BT2: the slope interpretation is what the sandwich would fix at the optimum	STRENGTHENED: the sandwich's ceiling is now the trace identity, so C awaits only the floor side; and the index itself is now a measured object on RockSample (E3)
BT1	the obstruction datum is isomorphic to the viability curvature	two-step chain: obstruction to Hankel structure [BT1a, open] to curvature [proved] to regulation [proved, with data]	UNCHANGED at the full-statement level: BT1a is the programme's sole remaining deep open link
BT2	$R(D)$ is the minimum per-optic Lipschitz constant of an optic implementing the transducer	the sandwich: the AAK floor below, the optic-chain ceiling above; equality at the optimum is the graded small-gain law, the open core	CLOSED AS FAR AS THE CORPUS ALLOWS: the ceiling is the guarded-trace identity; the gap law is exact (quadratic closed form); the equality locus is characterized; the floor inherits Open 7.13
BT3	the intercept principle is equivalent to non-existence of a fixed point of the homotopy fixed-point extension	enrichment first: once the cost-enriched graded category is built, the intercept becomes fixed-point non-existence of the currying endomorphism	DONE: the enrichment is built, the budget endomorphism is defined, and the intercept is PROVED as fixed-point non-existence (integer arithmetic), fused with the contraction side into the dichotomy law

Table 3 — The bridge ledger after this session: DeepSeek's six theorems, the corpus corrections of Volume II, and the status now

12 · What Remains

The record should close the way the audits have closed throughout: with the open items named, sized, and sequenced. The deep one is BT1a — the dictionary-plus-recovery theorem between the extensional diagnostics of causal descent and the operator structure of the Hankel theory: distinct rows as fibre conflicts, rank as interface size, the rank filtration as the nested-space ladder. Volume II called it the deep one because no existing template proves it, and nothing this session changes that; what this session changes is its isolation. With BT3 built and BT2 closed to the corpus's own conditional ceiling, BT1a is no longer one of three open links but the one, and the enrichment now exists to state it in: the obstruction datum is a section of the budget lattice's failure sheaf, and its identification with Hankel structure is a morphism problem in a category that finally has both sides.

The bounded ones are the floor and the benchmark's extensions. Open 7.13 — the intrinsic multiletter characterization of when the AAK floor is attainable — is now measured from the equality-locus side: the scan says the locus is thin, which is evidence about the characterization's shape but not the characterization. The unconditional small-gain law therefore still runs on the corpus's conditional floor, and says so wherever it is stated. On the benchmark side, the three bounded extensions carry their own sizes: Tiger and Hallway are small instances a single session can add; the production-solver comparison needs a machine with more than two cores and a C++ baseline; and the learning-efficiency clause needs the search-refinement result reversed at budgets two orders of magnitude up, which is where the literature says POMCP's tree stops depending on its default policy. The $n = 4$ scan's $L = 10$ leg, left to the manuscript's own computation in Volume IV's honest gaps, also still belongs on this list.

And the one that is not a theorem at all: the programme's unsubmitted state. Risk 4 was never a mathematical risk, and the session's work does not touch it; what it does is remove the last internal excuse on the two stages that precede it. The bridge paper's mathematics is written; the definitions exist; the law has its closed form and its measured walls; the benchmark has its first table and its honest negative datapoint. The order the user fixed for this session is executed in full — the enrichment built, the equality closed to its corpus-bounded ceiling, the benchmark confronted — and what the programme owes next is either BT1a's proof or the submission decision, whichever the author calls first. The mathematics will wait; the priority clock will not.

1 deep open link remaining: BT1a, the obstruction-to-Hankel dictionary-plus-recovery theorem	3 bounded extensions: Tiger and Hallway; the production-solver comparison; the learning-efficiency clause	0 internal excuses left on the bridge paper's mathematical readiness
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